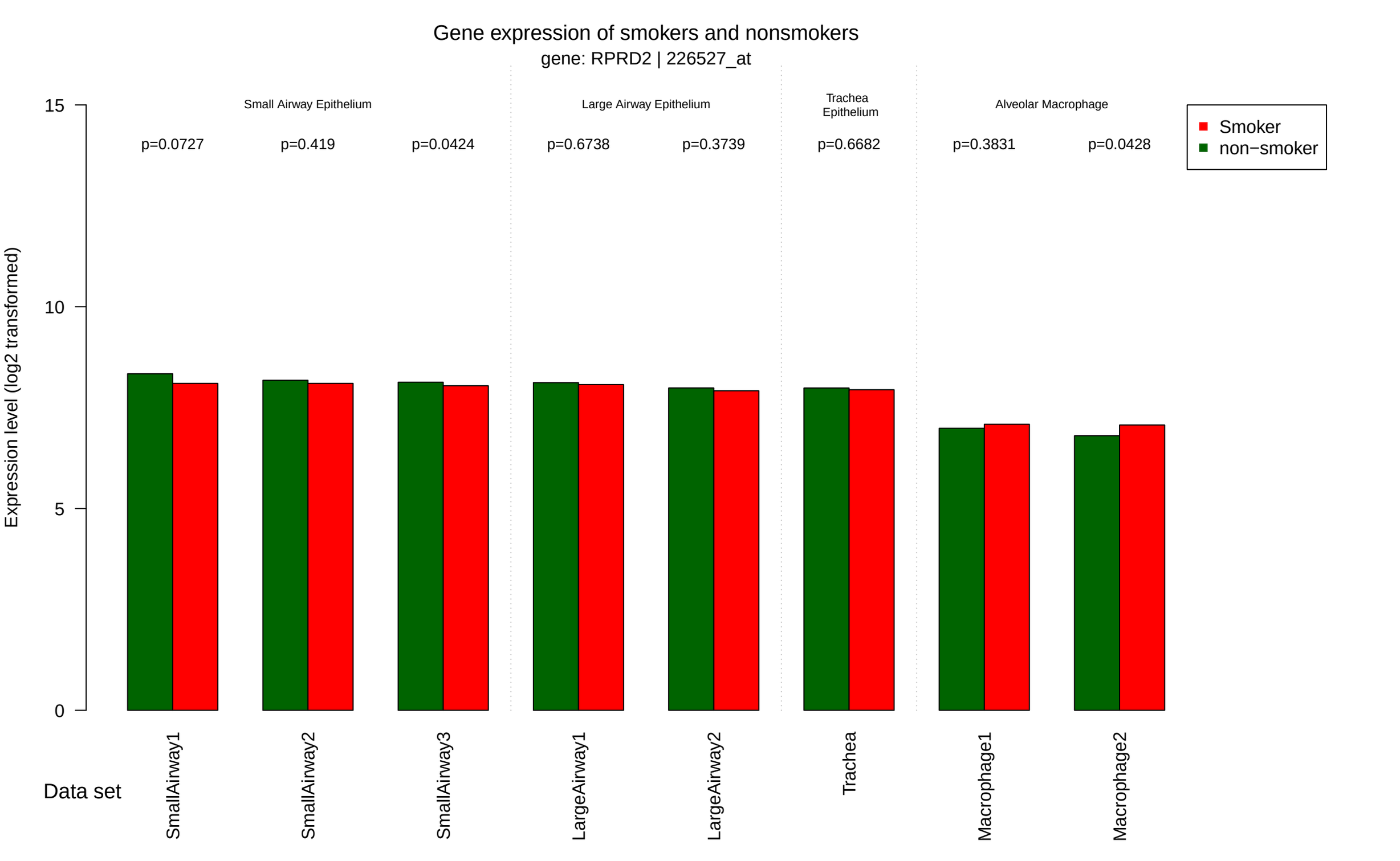
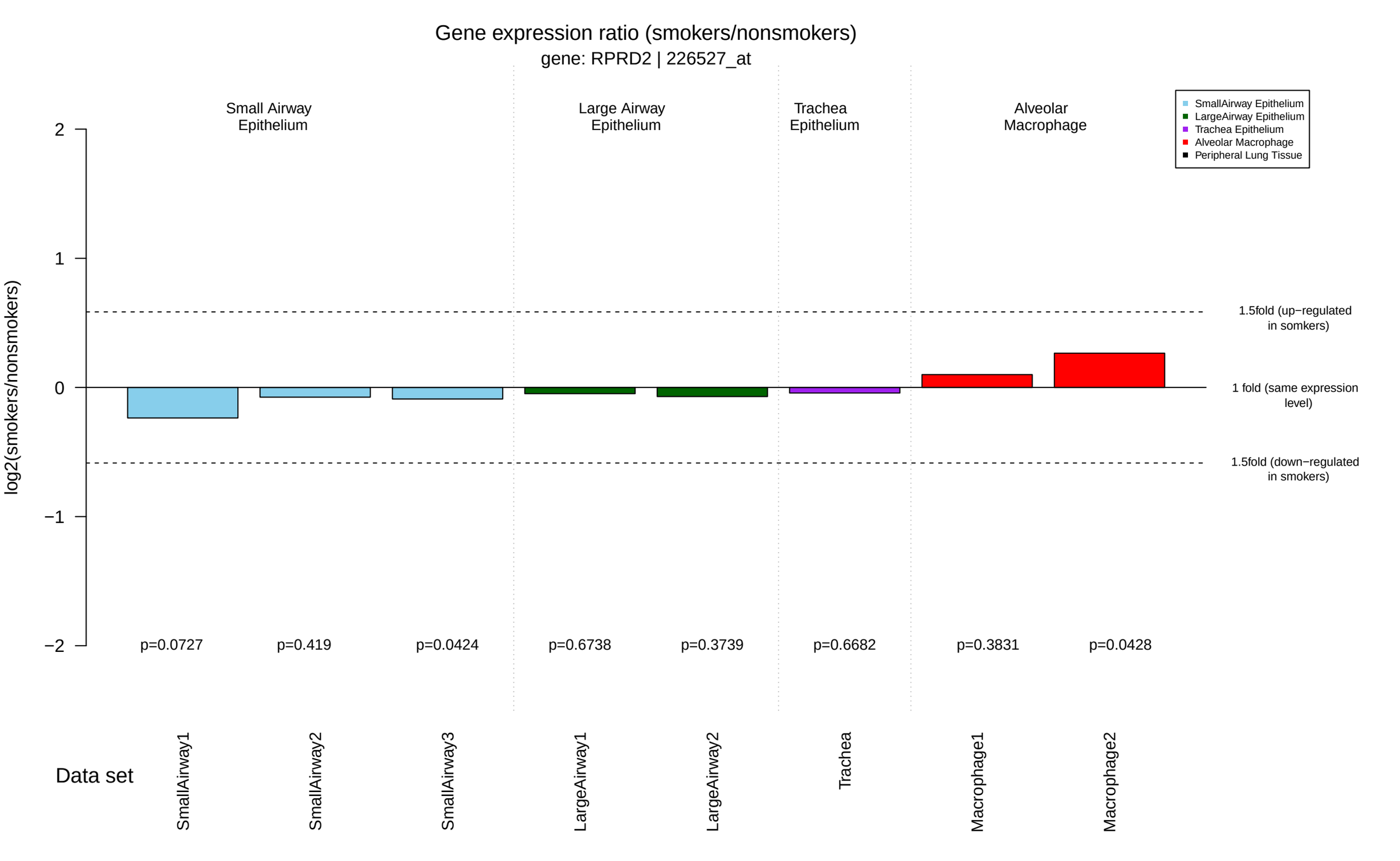
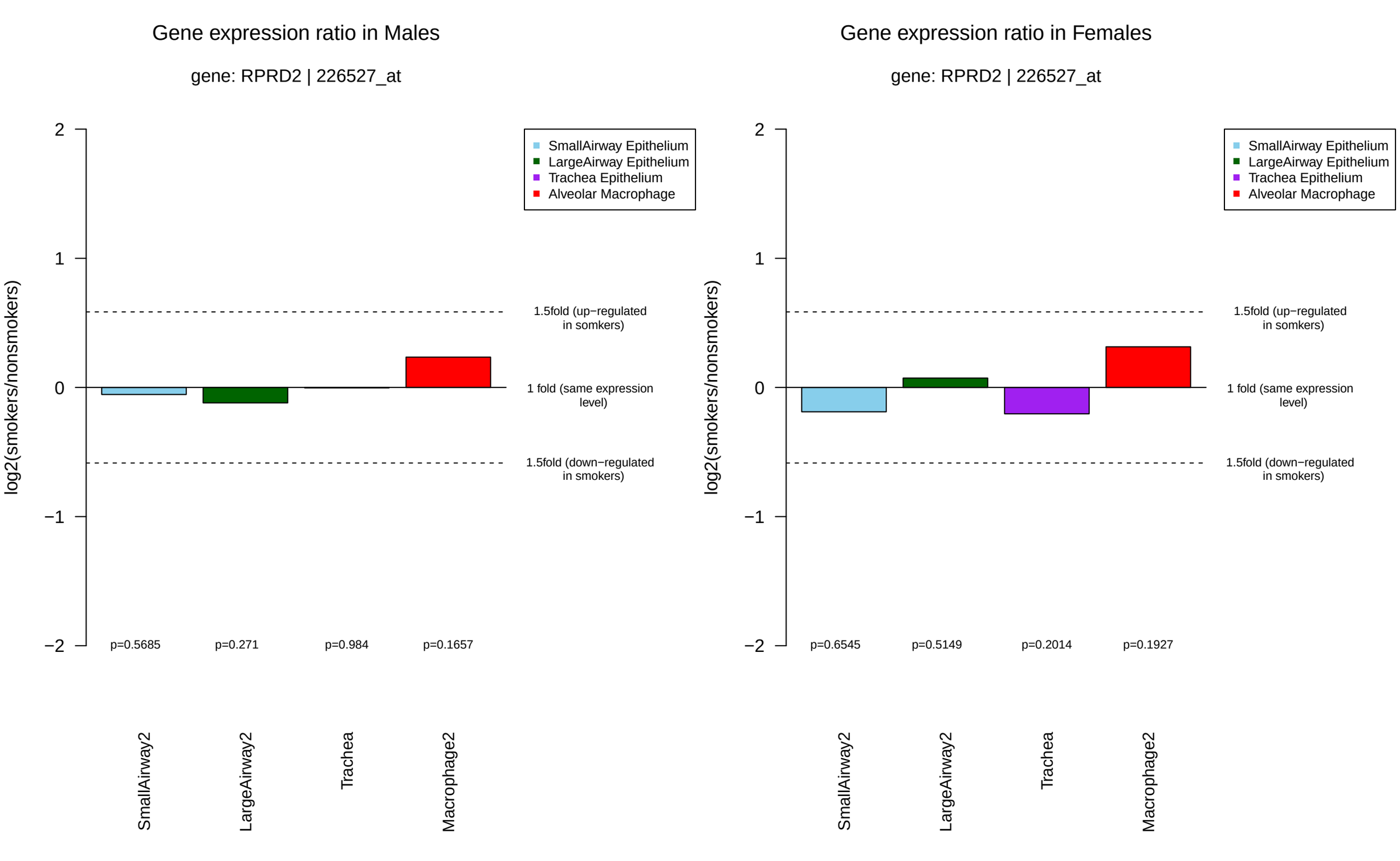
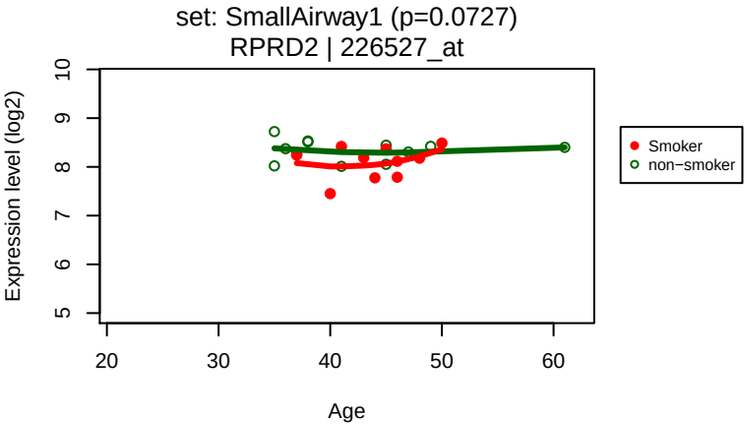
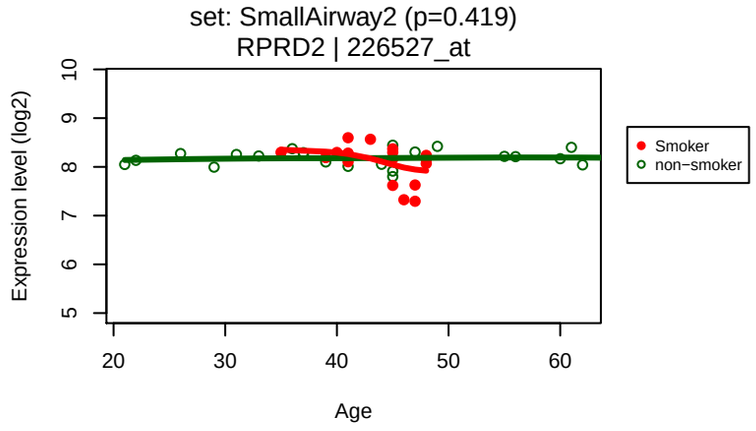
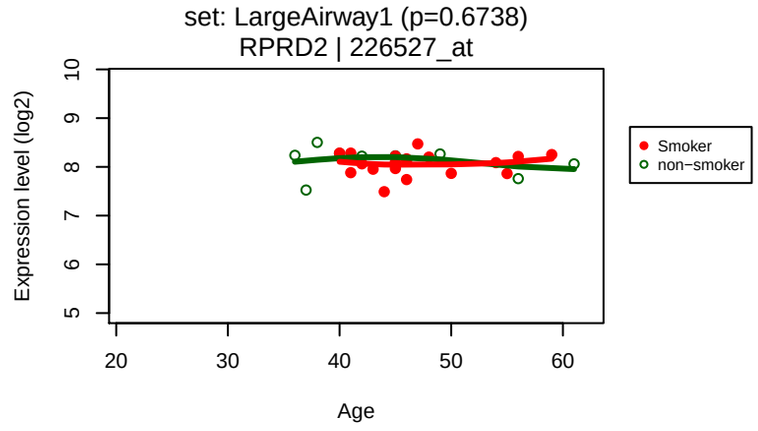

Gene expression of smokers and nonsmokers
gene: RPRD2 | 226527_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0727
p=0.419
p=0.0424
p=0.6738
p=0.3739
p=0.6682
p=0.3831
p=0.0428
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

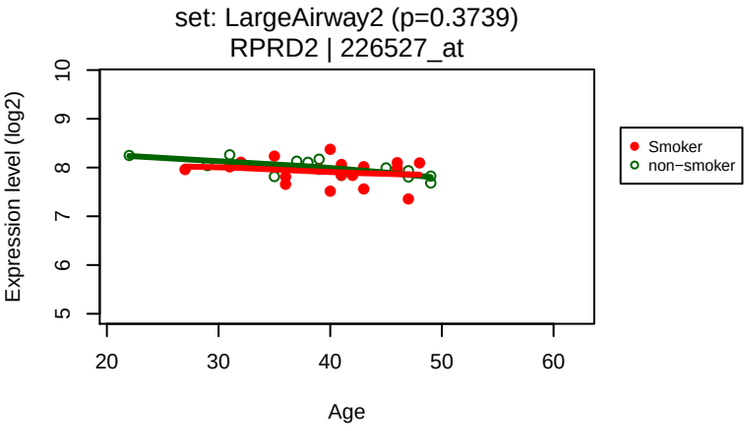

Gene expression ratio (smokers/nonsmokers)
gene: RPRD2 | 226527_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0727
p=0.419
p=0.0424
p=0.6738
p=0.3739
p=0.6682
p=0.3831
p=0.0428
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

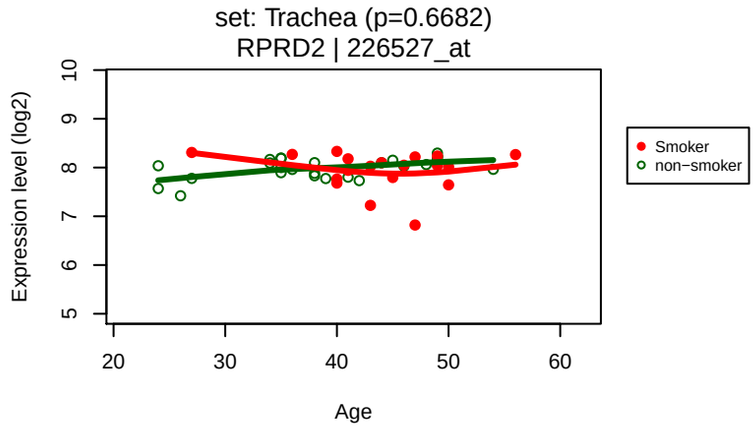

Gene expression ratio in Males
gene: RPRD2 | 226527_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5685
p=0.271
p=0.984
p=0.1657
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPRD2 | 226527_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6545
p=0.5149
p=0.2014
p=0.1927
SmallAirway2
LargeAirway2
Trachea
Macrophage2

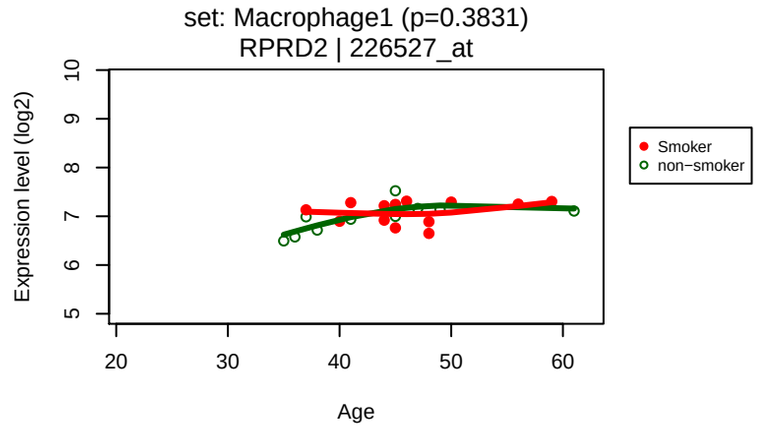

set: SmallAirway1 (p=0.0727)
RPRD2 | 226527_at
Expression level (log2)
Age
Smoker
non−smoker

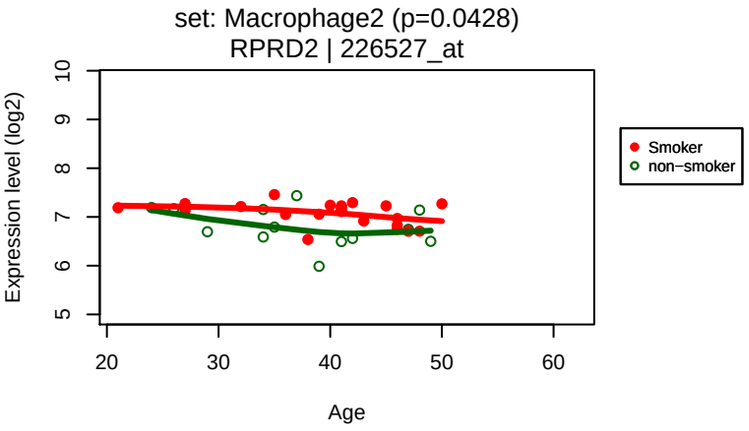

set: SmallAirway2 (p=0.419)
RPRD2 | 226527_at
Expression level (log2)
Age
Smoker
non−smoker

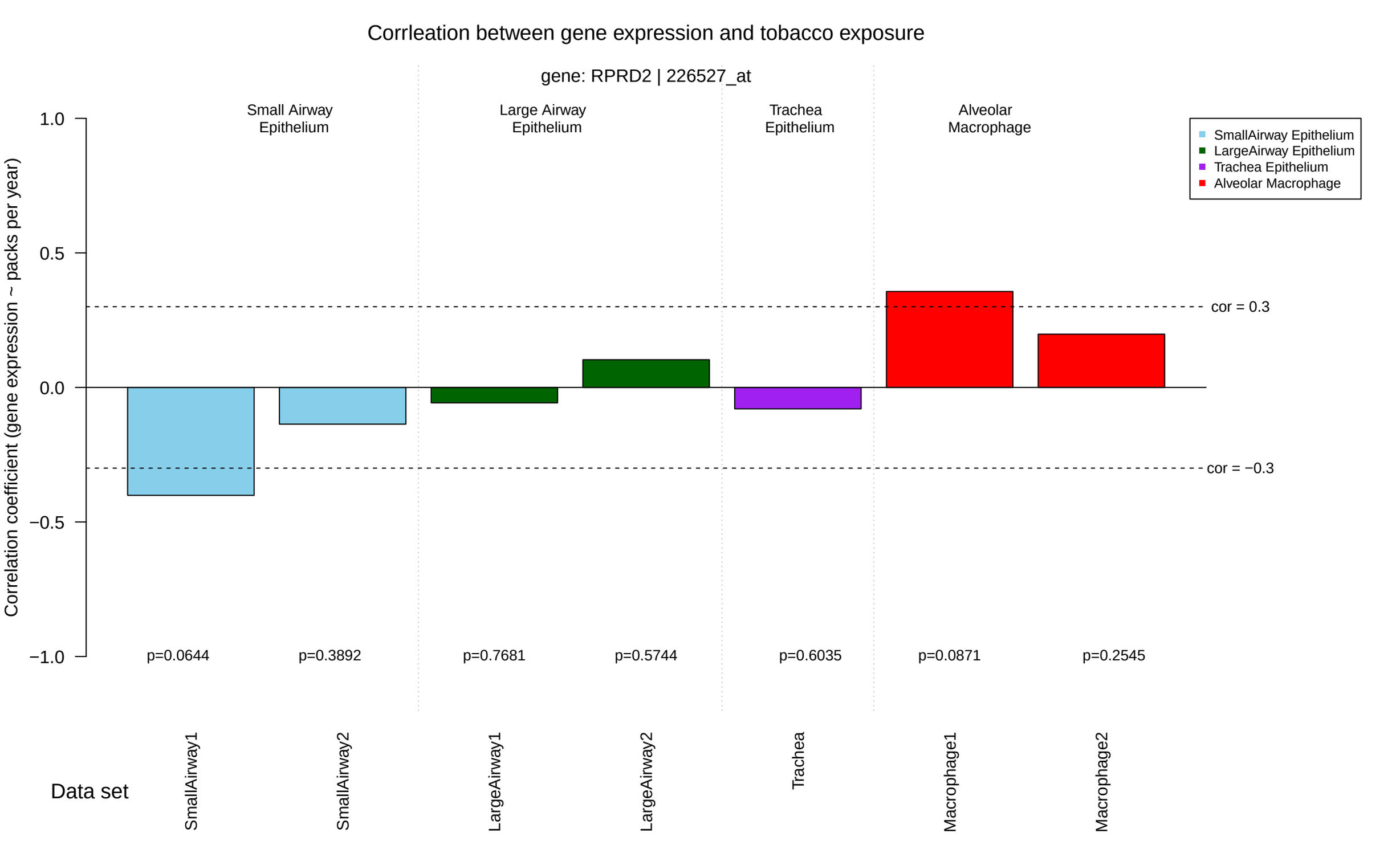

set: LargeAirway1 (p=0.6738)
RPRD2 | 226527_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3739)
RPRD2 | 226527_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6682)
RPRD2 | 226527_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3831)
RPRD2 | 226527_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0428)
RPRD2 | 226527_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPRD2 | 226527_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0644
p=0.3892
p=0.7681
p=0.5744
p=0.6035
p=0.0871
p=0.2545
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2